Database Relationships and Normalization

Superstore Central Region Analysis

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1. Database Design Overview

The Retail Sales & Profitability Analysis database was designed using a dimensional model with one central fact table and five dimension tables.

The FactSales table stores sales transactions and quantitative measures, including Sales, Profit, Quantity, and Discount. The dimension tables store descriptive information about customers, products, geography, orders, and dates.

The database was designed to organize the source data, reduce unnecessary repetition of descriptive information, and support efficient SQL-based analysis and reporting.

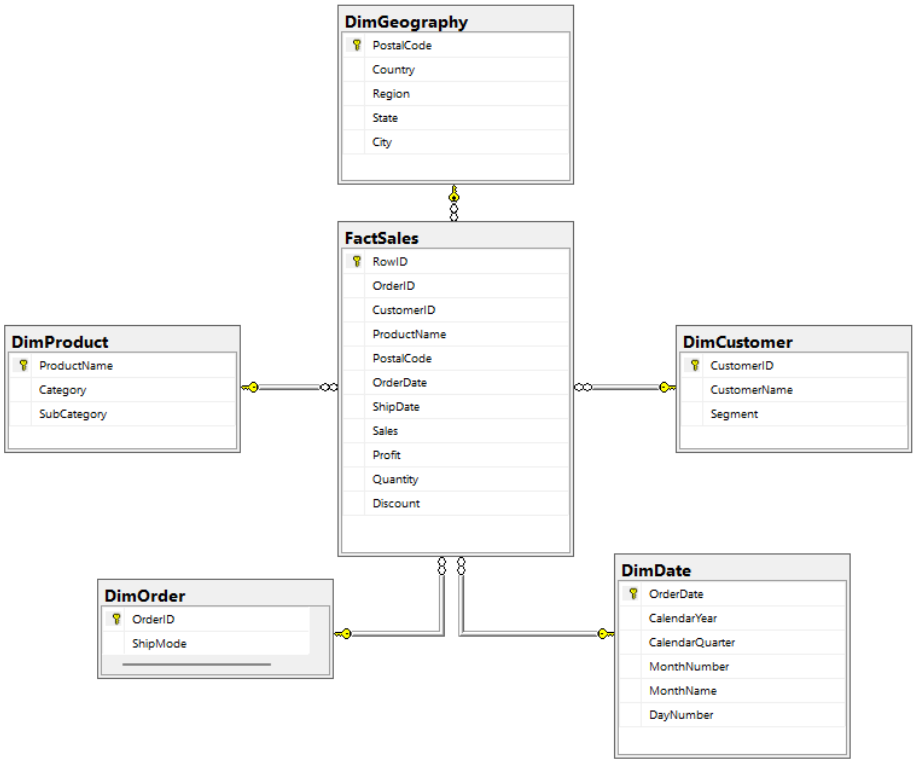

Figure 1. Database diagram: FactSales connected to five dimension tables

2. Relationships Between Tables

The FactSales table is connected to the five dimension tables through foreign key constraints.

- **DimCustomer** identifies the customer associated with each sales record.
- **DimProduct** provides descriptive information about the product.
- **DimGeography** provides geographical information, including state, city, and region.
- **DimOrder** stores order-level information, including the shipping mode.
- **DimDate** provides calendar attributes used for time-based analysis.

Each dimension is intended to have a one-to-many relationship with FactSales. A single dimension record can be associated with multiple sales records, while each sales record references its corresponding dimension records.

Foreign key constraints enforce referential integrity by requiring non-NULL foreign key values to match existing referenced values.

Dimension	Role	Relationship
DimCustomer	Identifies the customer on each sales record	1 : many
DimProduct	Descriptive product information	1 : many
DimGeography	State, city, and region	1 : many
DimOrder	Order-level data, including shipping mode	1 : many
DimDate	Calendar attributes for time-based analysis	1 : many

3. Normalization Summary

The source data was organized into separate dimension tables to avoid repeatedly storing customer, product, date, order, and geographical descriptions in every sales record.

The FactSales table retains the transactional measures needed for analysis, while the dimension tables provide descriptive attributes for grouping and filtering. Primary keys identify dimension records, and foreign keys establish the relationships between the fact and dimension tables.

This structure supports sales aggregation, profitability analysis, geographical comparisons, customer segmentation, and time-based reporting.